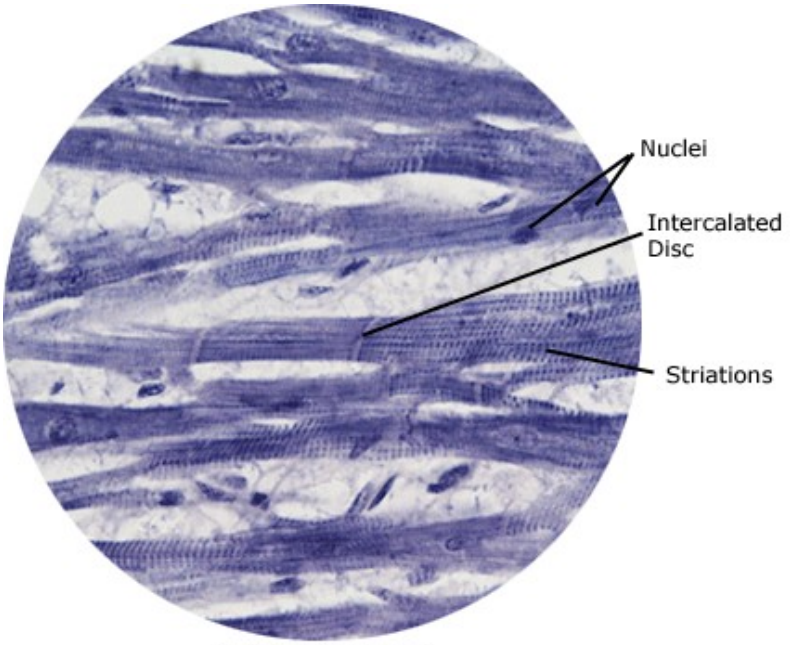
# *Cardiac Myocyte* *Prodigy Anesthesia*

A cardiac myocyte is a type of involuntary, striated muscle cell that is found primarily in the myocardium. Cardiac myocytes possess large quantities of mitochondria, making them resistant to fatigue. They possess low-resistance communication channels between one another called intercalated discs that allow for the quick and even spread of electrochemical signals.
While the action potentials of skeletal muscle cells is due to the abrupt opening of fast sodium channels, which creates the spike in the action potential, that of cardiac myocytes is due both to the opening of fast sodium channels AND slow calcium channels. The opening of the slower calcium channels creates the unique plateau following the spike in the cardiac action potential.



Cardiac Muscle Cell

*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 344-345.*

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